# Teaching Philosophy and Interests in Electrical and Computer Engineering

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## From an equation to a decision

A good engineering course should leave students able to explain not only a formula, but what it predicts, where its assumptions fail, and how to test it. When I teach a technical topic, I try to connect three activities: reason through the underlying model, implement or measure something tangible, and account for differences between prediction and observation. This approach matters in communication systems, where ideal channel models and real measurements seldom agree perfectly, but it also applies to signals, computing, and laboratory design.

I previously taught Computer Programming, Communication Systems, and Wireless Communication as a Lecturer in the University of Lahore's Department of Technology. At the University of Engineering and Technology, Taxila, I supported communications laboratory sessions, assignments, technical materials, and grading. At COMSATS University Islamabad, I helped undergraduate and graduate students with project and thesis research. I have also tutored tenth- and twelfth-grade students outside my university appointments. Working across levels of preparation has taught me how easily a missed prerequisite can turn into frustration, and how much progress students make when feedback identifies the next concrete step.

## Practical work that supports the theory

I favor teaching sequences that begin with a short explanation and worked example, followed by an exercise in which students must make and justify a choice. A lesson on sampling and frequency content, for example, could begin with hand calculations, continue with plots of a sampled waveform, and end with students explaining why an apparent low-frequency component may be an alias. Students could then modify a filter or sample rate in Python or MATLAB and compare their results against a prediction. In a communications laboratory, students could change a noise level or propagation parameter, record the observed impact, and explain which link-level conclusions are and are not supported by the experiment.

These are examples of learning activities I would develop, rather than a claim that I have previously delivered those exact laboratories. My documented instructor-of-record experience is in programming and communications. I have technical preparation in signal processing, networked systems, engineering computation, and machine learning; when assigned a foundational course outside my prior teaching record, I would prepare against its prerequisites, approved syllabus, laboratory equipment, and faculty guidance. I would not portray my wireless research as experience teaching power electronics, integrated-circuit fabrication, PLCs, or VLSI.

My industry work with AT&T Labs, Futurewei, and AT&T provides useful examples of imperfect measurements, changing conditions, and engineering tradeoffs. Where appropriate, I would use anonymized or openly available examples rather than proprietary records. Students should learn that a working simulation is a step toward engineering judgment, not proof that a proposed design will behave identically in a deployed system.

## Assessment, feedback, and support of diverse learners

I start with explicit learning outcomes: derive a relationship, read a signal or a program, design a test, evaluate uncertainty, or explain a tradeoff. I would use low-stakes questions and brief checkpoints to surface misunderstandings before a major assignment or examination. For design and lab projects, a rubric should distinguish conceptual understanding, implementation, testing, interpretation, and communication. A student who documents a failed experiment carefully can demonstrate more engineering learning than one who produces an attractive result without knowing why it occurred.

Supporting students from varied educational and personal backgrounds is an everyday teaching responsibility. I would make software and laboratory prerequisites visible, provide a tested starting environment and clear instructions, and offer alternative explanations when a concept is not landing. Structured group roles can prevent programming or equipment operation from falling repeatedly to the student who arrived with the most experience. Office hours, transparent rubrics,

accessible materials, and timely formative feedback give students routes to meet demanding standards without assuming identical preparation. I would follow institutional accessibility procedures and seek colleagues' input when a course needs stronger support structures.

Students also benefit from seeing responsible engineering practice modeled in class. I would invite discussion of measurement quality, energy or computation cost, privacy when sensor data are involved, and the consequences of relying on an untested model. These conversations should deepen the core engineering lesson, not replace it.

### Project mentoring and continuous improvement

I would guide undergraduate projects through a sequence of achievable milestones: a question and requirements, an established baseline, a test plan, a prototype, an evaluation, and a concise report. Students should learn to document assumptions and version their work. In a senior design setting, I would ask teams to divide implementation, testing, validation, and presentation responsibilities, with individual accountability and regular progress reviews.

I see course improvement as an experimental process as well. Patterns in student submissions, short midterm feedback, and discussion with department colleagues can reveal where an explanation or assignment is not doing its job. I want students to leave my courses with sound technical foundations, the confidence to ask good questions, and the discipline to test answers before trusting them.